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FIG. 3

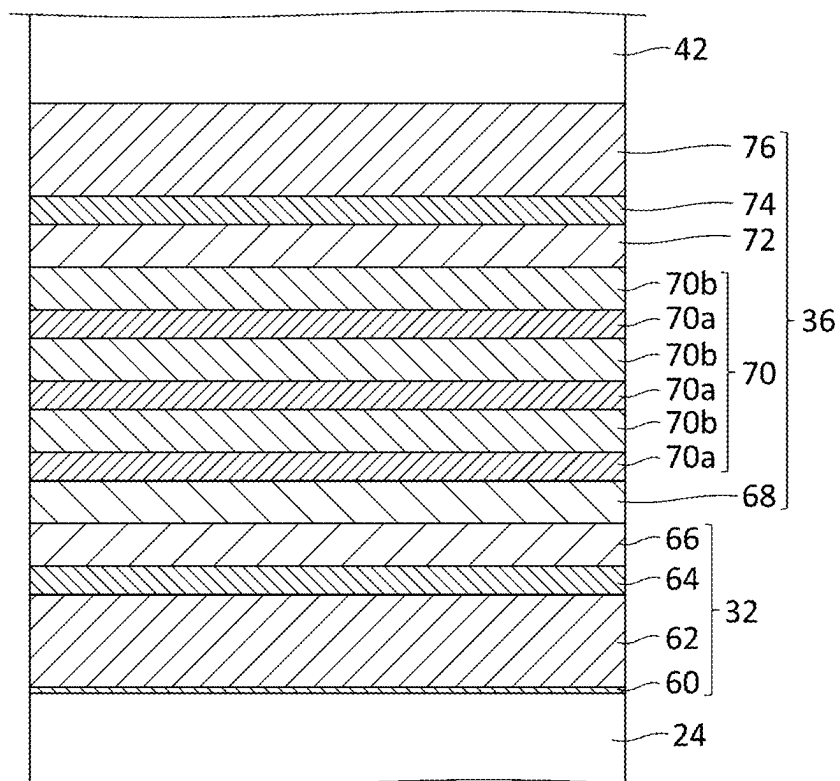


FIG. 4

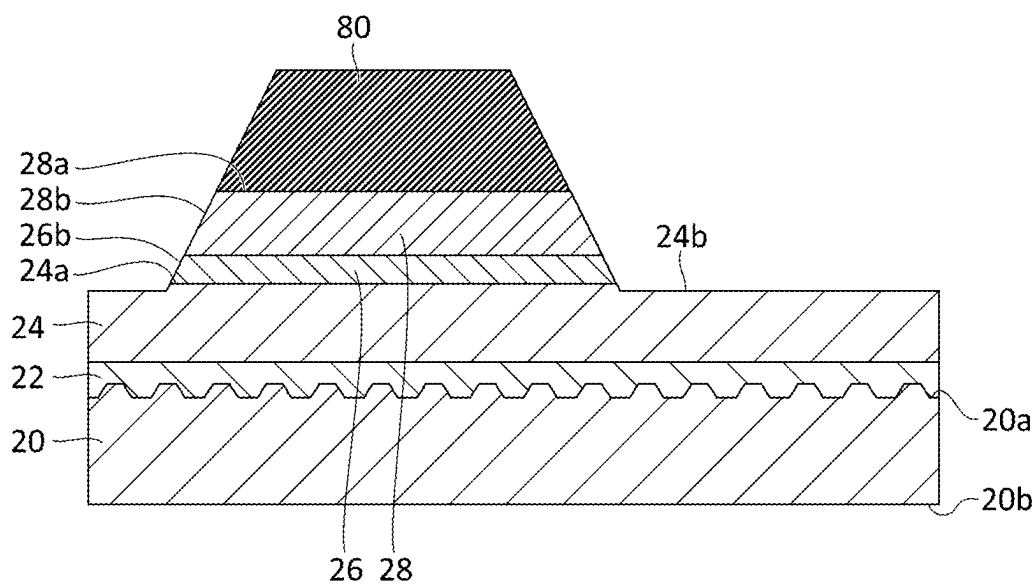


FIG. 5

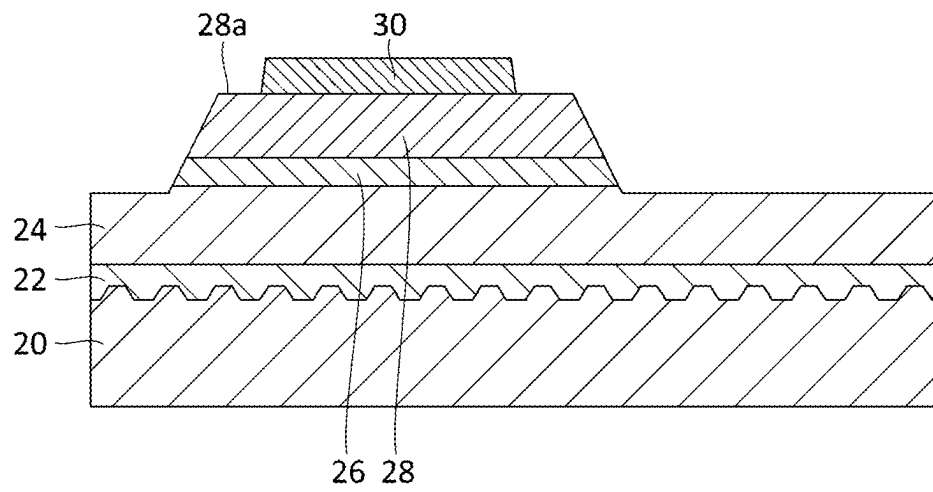


FIG. 6

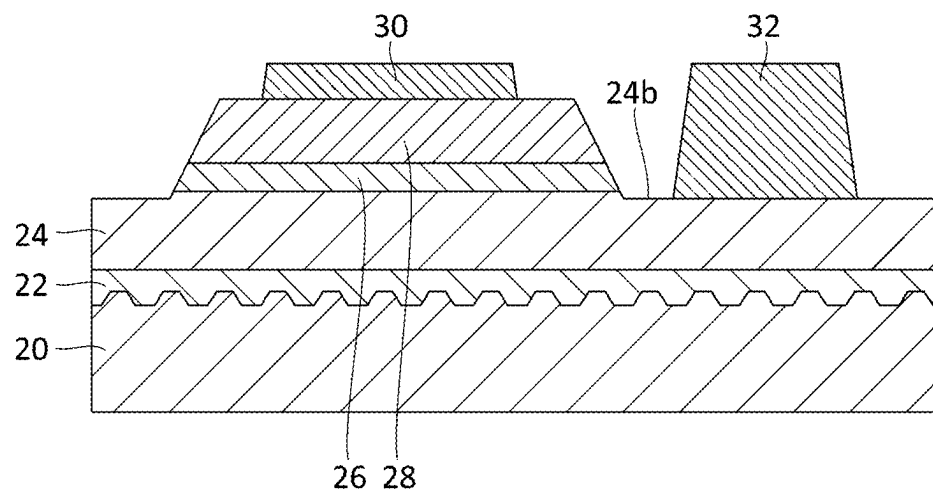


FIG. 7

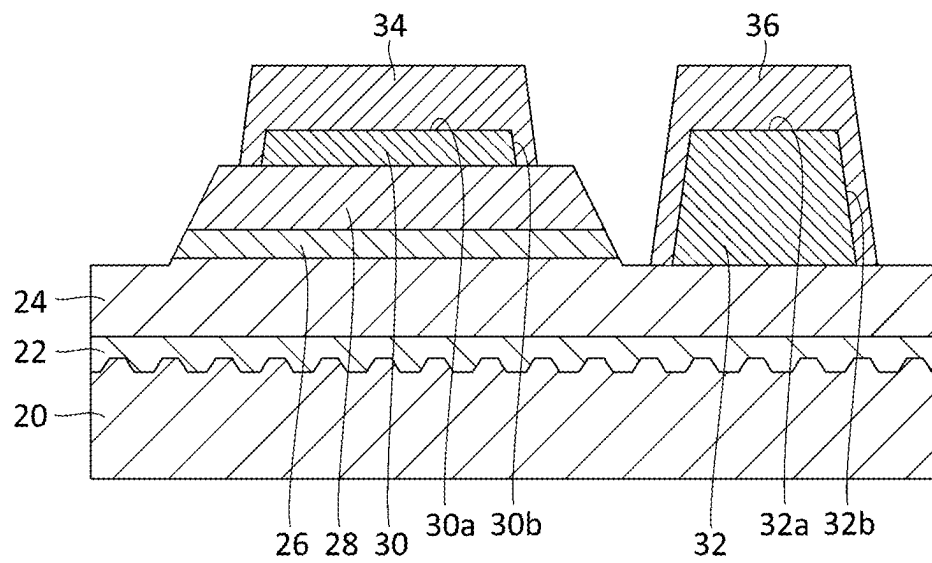


FIG. 8

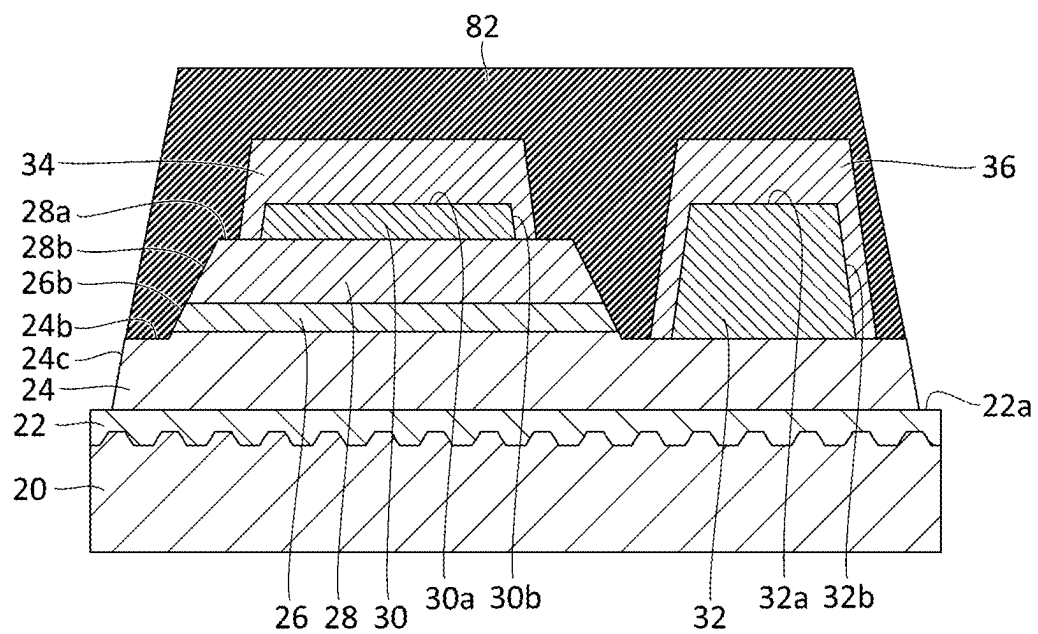
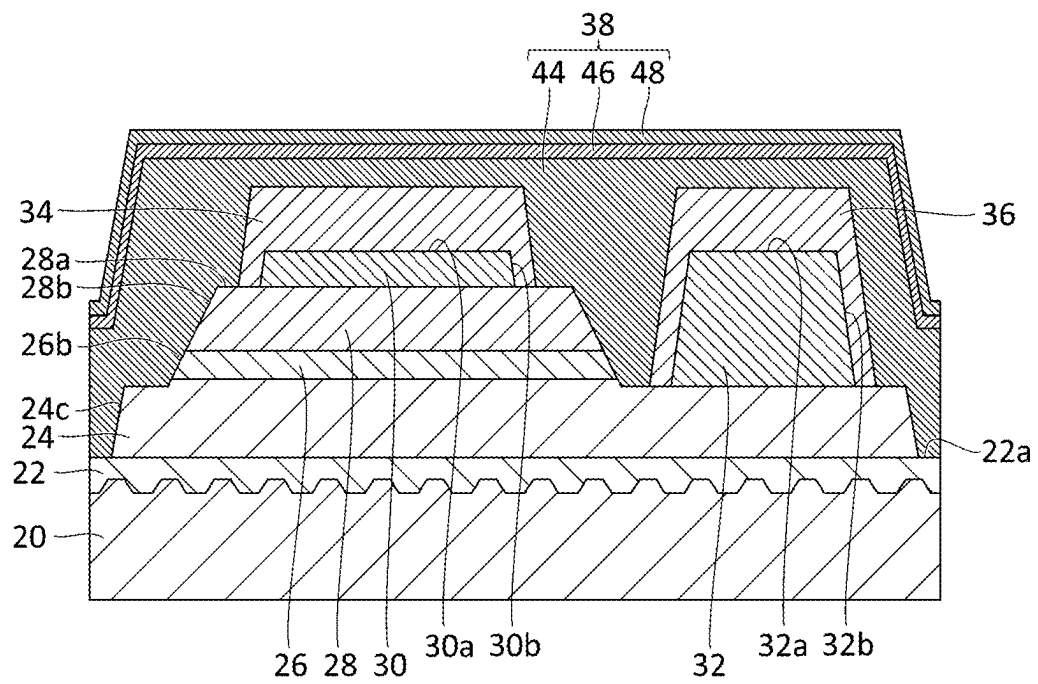


FIG. 9



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SEMICONDUCTOR LIGHT-EMITTING ELEMENT

RELATED APPLICATION

Priority is claimed to Japanese Patent Application No. 2021-126107, filed on Jul. 30, 2021, the entire content of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to a semiconductor light-emitting element.

2. Description of the Related Art

A semiconductor light-emitting element includes an n-type semiconductor layer, an active layer, and a p-type semiconductor layer stacked on a substrate. A p-side contact electrode is provided on the p-type semiconductor layer, and a dielectric layer is provided on the p-side contact electrode. In a semiconductor light-emitting element for outputting deep ultraviolet light, rhodium (Rh) having a high reflectivity for the emission wavelength may be used in the p-side contact electrode (see, for example, JP6839320B).

Adhesion of Rh to a dielectric material is poor so that the element reliability may be lowered due to exfoliation of the dielectric layer. Further, when the dielectric layer on the p-side contact electrode is removed to form a connection opening, a damage associated with the removal of the dielectric layer may be caused in the p-side contact electrode exposed in the connection opening with the result that the reflective characteristics of the p-side contact electrode may be lowered.

SUMMARY OF THE INVENTION

The present invention addresses the above-described issues and a purpose thereof is to provide a technology for improving the reliability of a semiconductor light-emitting element.

A semiconductor light-emitting element according to an embodiment of the present invention includes: an n-type semiconductor layer made of an n-type AlGaIn-based semiconductor material; an active layer provided on the n-type semiconductor layer and made of an AlGaIn-based semiconductor material; a p-type semiconductor layer provided on the active layer; a p-side contact electrode that includes a Rh layer in contact with an upper surface of the p-type semiconductor layer; and a p-side current diffusion layer that is in contact with an upper surface and a side surface of the p-side contact electrode and includes a TiN layer, a Ti layer, a Rh layer, and a TiN layer stacked successively. An Ar concentration in the Rh layer included in the p-side contact electrode is smaller than an Ar concentration in the Rh layer included in the p-side current diffusion layer.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a cross sectional view schematically showing a configuration of a semiconductor light-emitting element according to an embodiment;

FIG. 2 is a cross sectional view schematically showing a configuration of the p-side contact electrode and the p-side current diffusion layer;

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FIG. 3 is a cross sectional view schematically showing a configuration of the n-side contact electrode and the n-side current diffusion layer;

FIG. 4 schematically shows a step of manufacturing the semiconductor light-emitting element;

FIG. 5 schematically shows a step of manufacturing the semiconductor light-emitting element;

FIG. 6 schematically shows a step of manufacturing the semiconductor light-emitting element;

FIG. 7 schematically shows a step of manufacturing the semiconductor light-emitting element;

FIG. 8 schematically shows a step of manufacturing the semiconductor light-emitting element; and

FIG. 9 schematically shows a step of manufacturing the semiconductor light-emitting element.

DETAILED DESCRIPTION OF THE INVENTION

The invention will now be described by reference to the preferred embodiments. This does not intend to limit the scope of the present invention, but to exemplify the invention.

A detailed description will be given of an embodiment of the present invention with reference to the drawings. The numerals are used in the description to denote the same elements and a duplicate description is omitted as appropriate. To facilitate the understanding, the relative dimensions of the constituting elements in the drawings do not necessarily mirror the relative dimensions in the actual light-emitting element.

The semiconductor light-emitting element according to the embodiment is configured to emit “deep ultraviolet light” having a central wavelength λ approximately equal to or less than 360 nm and is a so-called deep ultraviolet-light-emitting diode (UV-LED) chip. To output deep ultraviolet light having such a wavelength, an aluminum gallium nitride (AlGaIn)-based semiconductor material having a band gap approximately equal to or more than 3.4 eV is used. The embodiment particularly shows a case of emitting deep ultraviolet light having a central wavelength λ of about 240 nm-320 nm.

In this specification, the term “AlGaIn-based semiconductor material” refers to a semiconductor material containing at least aluminum nitride (AlN) and gallium nitride (GaIn) and shall encompass a semiconductor material containing other materials such as indium nitride (InN). Therefore, “AlGaIn-based semiconductor materials” as recited in this specification can be represented by a composition $\text{In}_{1-x-y}\text{Al}_x\text{Ga}_y\text{N}$ ($0 < x + y \leq 1$, $0 < x < 1$, $0 < y < 1$). The AlGaIn-based semiconductor material shall encompass AlGaIn or InAlGaIn. The “AlGaIn-based semiconductor material” in this specification has a molar fraction of AlN and a molar fraction of GaIn equal to or more than 1%, and, preferably, equal to or more than 5%, equal to or more than 10%, or equal to or more than 20%.

Those materials that do not contain AlN may be distinguished by referring to them as “GaIn-based semiconductor materials”. “GaIn-based semiconductor materials” include GaIn or InGaIn. Similarly, those materials that do not contain GaIn may be distinguished by referring to them as “AlN-based semiconductor materials”. “AlN-based semiconductor materials” include AlN or InAlN.

FIG. 1 is a cross sectional view schematically showing a configuration of a semiconductor light-emitting element 10 according to the embodiment. The semiconductor light-emitting element 10 includes a substrate 20, a base layer 22,

an n-type semiconductor layer **24**, an active layer **26**, a p-type semiconductor layer **28**, a p-side contact electrode **30**, an n-side contact electrode **32**, a p-side current diffusion layer **34**, an n-side current diffusion layer **36**, a protective layer **38**, a p-side pad electrode **40**, and an n-side pad electrode **42**.

Referring to FIG. 1, the direction indicated by the arrow A may be referred to as “vertical direction” or “direction of thickness”. Further, the direction away from the substrate **20** may be defined as “upward”, and the direction toward the substrate **20** may be defined as “downward”.

The substrate **20** includes a first principal surface **20a** and a second principal surface **20b** opposite to the first principal surface **20a**. The first principal surface **20a** is a crystal growth surface for growing the layers from the base layer **22** to the p-type semiconductor layer **28**. The substrate **20** is made of a material having translucency for the deep ultraviolet light emitted by the semiconductor light-emitting element **10** and is made of, for example, sapphire (Al_2O_3). A fine concave-convex pattern having a submicron (1 μm or less) depth and pitch is formed on the first principal surface **20a**. The substrate **20** like this is also called a patterned sapphire substrate (PSS). The second principal surface **20b** is a light extraction substrate for extracting the deep ultraviolet light emitted by the active layer **26** outside. The substrate **20** may be made of AlN or made of AlGaIn. The substrate **20** may be an ordinary substrate in which the first principal surface **20a** is comprised of a flat surface that is not patterned.

The base layer **22** is provided on the first principal surface **20a** of the substrate **20**. The base layer **22** is a foundation layer (template layer) to form the n-type semiconductor layer **24**. For example, the base layer **22** is an undoped AlN layer and is, specifically, an AlN (HT-AlN; High Temperature AlN) layer grown at a high temperature. The base layer **22** may further include an undoped AlGaIn layer formed on the AlN layer. The base layer **22** may be comprised only of an undoped AlGaIn layer when the substrate **20** is an AlN substrate or an AlGaIn substrate. In other words, the base layer **22** includes at least one of an undoped AlN layer or an undoped AlGaIn layer.

The n-type semiconductor layer **24** is provided on an upper surface **22a** of the base layer **22**. The n-type semiconductor layer **24** is made of an n-type AlGaIn-based semiconductor material. For example, the n-type semiconductor layer **24** is doped with Si as an n-type impurity. The composition ratio of the n-type semiconductor layer **24** is selected to transmit the deep ultraviolet light emitted by the active layer **26**. For example, the n-type semiconductor layer **24** is formed such that the molar fraction of AlN is equal to or more than 25%, and, preferably, equal to or more than 40% or equal to or more than 50%. The n-type semiconductor layer **24** has a band gap larger than the wavelength of the deep ultraviolet light emitted by the active layer **26**. For example, the n-type semiconductor layer **24** is formed to have a band gap equal to or more than 4.3 eV. It is preferable to form the n-type semiconductor layer **24** such that the molar fraction of AlN is equal to or less than 80%, i.e., the band gap is equal to or less than 5.5 eV. It is more desired to form the n-type semiconductor layer **24** such that the molar fraction of AlN is equal to or less than 70% (i.e., the band gap is equal to or less than 5.2 eV). The n-type semiconductor layer **24** has a thickness equal to or more than 1 μm and equal to or less than 3 μm . For example, the n-type semiconductor layer **24** has a thickness of about 2 μm .

The n-type semiconductor layer **24** is formed such that the concentration of Si as the impurity is equal to or more than

$1 \times 10^{18}/\text{cm}^3$ and equal to or less than $5 \times 10^{19}/\text{cm}^3$. It is preferred to form the n-type semiconductor layer **24** such that the Si concentration is equal to or more than $5 \times 10^{18}/\text{cm}^3$ and equal to or less than $3 \times 10^{19}/\text{cm}^3$ and, more preferably, equal to or more than $7 \times 10^{18}/\text{cm}^3$ and equal to or less than $2 \times 10^{19}/\text{cm}^3$. In one example, the Si concentration in the n-type semiconductor layer **24** is around $1 \times 10^{19}/\text{cm}^3$ and, more specifically, is in a range equal to or more than $8 \times 10^{18}/\text{cm}^3$ and equal to or less than $1.5 \times 10^{19}/\text{cm}^3$.

The n-type semiconductor layer **24** includes a first upper surface **24a** and a second upper surface **24b**. The first upper surface **24a** is where the active layer **26** is formed, and the second upper surface **24b** is where the active layer **26** is not formed.

The active layer **26** is provided on the first upper surface **24a** of the n-type semiconductor layer **24**. The active layer **26** is made of an AlGaIn-based semiconductor material and has a double heterojunction structure by being sandwiched between the n-type semiconductor layer **24** and the p-type semiconductor layer **28**. To output deep ultraviolet light having a wavelength equal to or less than 355 nm, the active layer **26** is formed to have a band gap equal to or more than 3.4 eV. For example, the AlN composition ratio of the active layer **26** is selected so as to output deep ultraviolet light having a wavelength equal to or less than 320 nm.

For example, the active layer **26** has a monolayer or multilayer quantum well structure and is comprised of a barrier layer made of an undoped AlGaIn-based semiconductor material and a well layer made of an undoped AlGaIn-based semiconductor material. The active layer **26** includes, for example, a first barrier layer directly in contact with the n-type semiconductor layer **24** and a first well layer provided on the first barrier layer. One or more pairs of the barrier layer and the well layer may be additionally provided between the first well layer and the p-type semiconductor layer **28**. Each of the barrier layer and the well layer has a thickness equal to or more than 1 nm and equal to or less than 20 nm, and have, for example, a thickness equal to or more than 2 nm and equal to or less than 10 nm.

An electron blocking layer may further be provided between the active layer **26** and the p-type semiconductor layer **28**. The electron blocking layer is made of an undoped AlGaIn-based semiconductor material and is formed such that, for example, the molar fraction of AlN is equal to or more than 40%, and, preferably, equal to or more than 50%. The electron blocking layer may be formed such that the molar fraction of AlN is equal to or more than 80% or may be made of an AlN-based semiconductor material that does not contain GaN. The electron blocking layer has a thickness equal to or more than 1 nm and equal to or less than 10 nm. For example, the electron blocking layer has a thickness equal to or more than 2 nm and equal to or less than 5 nm.

The p-type semiconductor layer **28** is formed on the active layer **26**. The p-type semiconductor layer **28** is a p-type AlGaIn-based semiconductor material layer or a p-type GaN-based semiconductor material layer. For example, the p-type semiconductor layer **28** is an AlGaIn layer or a GaIn layer doped with magnesium (Mg) as a p-type impurity. The p-type semiconductor layer **28** has, for example, a thickness equal to or more than 20 nm and equal to or less than 400 nm.

The p-type semiconductor layer **28** may be comprised of a plurality of layers. The p-type semiconductor layer **28** may include, for example, a p-type clad layer and a p-type contact layer. The p-type clad layer is a p-type AlGaIn layer having a relatively high AlN ratio as compared with the p-type contact layer and is provided to be directly in contact with

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the active layer 26. The p-type contact layer is a p-type AlGaIn layer or a p-type GaN layer having a relatively low AlN ratio as compared with the p-type clad layer. The p-type contact layer is provided on the p-type clad layer and is provided to be directly in contact with the p-side contact electrode 30. The p-type clad layer may include a p-type first clad layer and a p-type second clad layer.

The composition ratio of the p-type first clad layer is selected to transmit the deep ultraviolet light emitted by the active layer 26. For example, the p-type first clad layer is formed such that the molar fraction of AlN is equal to or more than 25%, and, preferably, equal to or more than 40% or equal to or more than 50%. The AlN ratio of the p-type first clad layer is, for example, similar to the AlN ratio of the n-type semiconductor layer 24 or larger than the AlN ratio of the n-type semiconductor layer 24. The AlN ratio of the p-type clad layer may be equal to or more than 70% or equal to or more than 80%. The p-type first clad layer has a thickness equal to or more than 10 nm and equal to or less than 100 nm. For example, the p-type first clad layer has a thickness equal to or more than 15 nm and equal to or less than 70 nm.

The p-type second clad layer is provided on the p-type first clad layer. The p-type second clad layer is a p-type AlGaIn layer having a medium AlN ratio and has an AlN ratio lower than that of the p-type first clad layer and higher than that of the p-type contact layer. For example, the p-type second clad layer is formed such that the molar fraction of AlN is equal to or more than 25%, and, preferably, equal to or more than 40% or equal to or more than 50%. The AlN ratio of the p-type second clad layer is configured to be, for example, about $\pm 10\%$ of the AlN ratio of the n-type semiconductor layer 24. The p-type second clad layer has a thickness equal to or more than 5 nm and equal to or less than 250 nm and has, for example, a thickness equal to or more than 10 nm and equal to or less than 150 nm. The p-type second clad layer may not be provided. The p-type clad layer may be comprised only of the p-type first clad layer.

The p-type contact layer is a p-type AlGaIn layer or a p-type GaN layer having a relatively low AlN ratio. The p-type contact layer is formed such that the AlN ratio is equal to or less than 20% in order to obtain proper ohmic contact with the p-side contact electrode 30. Preferably, the p-type contact layer is formed such that the AlN ratio is equal to or less than 10%, equal to or less than 5%, or 0%. In other words, the p-type contact layer may be made of a p-type GaN-based semiconductor material that does not substantially contain AlN. As a result, the p-type contact layer could absorb the deep ultraviolet light emitted by the active layer 26. It is preferred to form the p-type contact layer to be thin to reduce the quantity of absorption of the deep ultraviolet light emitted by the active layer 26. The p-type contact layer has a thickness equal to or more than 5 nm and equal to or less than 30 nm and has, for example, a thickness equal to or more than 10 nm and equal to or less than 20 nm.

The p-side contact electrode 30 is provided on an upper surface 28a of the p-type semiconductor layer 28. The p-side contact electrode 30 can be in ohmic contact with the p-type semiconductor layer 28 (for example, the p-type contact layer) and is made of a material having a high reflectivity for the deep ultraviolet light emitted by the active layer 26. The p-side contact electrode 30 includes a Rh layer directly in contact with the upper surface 28a of the p-type semiconductor layer 28. The p-side contact electrode 30 may be, for example, comprised only of the Rh layer. The thickness of

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the Rh layer included in the p-side contact electrode 30 is equal to or more than 50 nm and equal to or less than 200 nm and is, for example, equal to or more than 70 nm and equal to or less than 150 nm.

The film density of the Rh layer included in the p-side contact electrode 30 is equal to or more than 12.0 g/cm^3 and is, for example, equal to or more than 12.2 g/cm^3 and equal to or less than 12.5 g/cm^3 . By configuring the film density of the Rh layer included in the p-side contact electrode 30 to be large, the function of the p-side contact electrode 30 as a reflection electrode can be enhanced. By configuring the film density of the Rh layer to be equal to or more than 12 g/cm^3 , the reflectivity equal to or more than 65% for deep ultraviolet light having a wavelength of 280 nm can be obtained. By way of one example, the Rh layer included in the p-side contact electrode 30 has a film density of 12.42 g/cm^3 and has a reflectivity of 66.8% for deep ultraviolet light having a wavelength of 280 nm. Further, the Ar concentration in the Rh layer included in the p-side contact electrode 30 is equal to or more than $1 \times 10^{16} \text{ /cm}^3$ and less than $1 \times 10^{18} \text{ /cm}^3$. By configuring the Ar concentration in the Rh layer included in the p-side contact electrode 30 to be low, the film quality of the Rh layer can be improved and the function of the p-side contact electrode 30 as a reflection electrode can be enhanced.

The n-side contact electrode 32 is provided on the second upper surface 24b of the n-type semiconductor layer 24. The n-side contact electrode 32 includes a first Ti layer, an Al layer, a second Ti layer, and a TiN layer stacked successively. The detail of the configuration of the n-side contact electrode 32 will be described separately with reference to FIG. 3.

The p-side current diffusion layer 34 is provided to be directly in contact with an upper surface 30a and a side surface 30b of the p-side contact electrode 30 and to cover the entirety of the p-side contact electrode 30. The p-side current diffusion layer 34 includes a TiN layer, a Ti layer, a Rh layer, and a TiN layer stacked successively. The detail of the configuration of the p-side current diffusion layer 34 will be described separately with reference to FIG. 2.

The n-side current diffusion layer 36 is provided to cover an upper surface 32a and a side surface 32b of the n-side contact electrode 32. The n-side current diffusion layer 36 is configured in a manner similar to the p-side current diffusion layer 34 and includes a TiN layer, a Ti layer, a Rh layer, and a TiN layer stacked successively. The detail of the configuration of the n-side current diffusion layer 36 will be described separately with reference to FIG. 3.

The protective layer 38 includes a p-side pad opening 38p provided on the p-side current diffusion layer 34 and an n-side pad opening 38n provided on the n-side current diffusion layer 36. The protective layer 38 covers the p-side current diffusion layer 34 in a portion different from the p-side pad opening 38p and covers the n-side current diffusion layer 36 in a portion different from the n-side pad opening 38n. The protective layer 38 is provided to cover the entirety of the element from above and is provided to cover the upper surface 22a of the base layer 22, the upper surface 24b and a side surface 24c of the n-type semiconductor layer 24, a side surface 26b of the active layer 26, the upper surface 28a and a side surface 28b of the p-type semiconductor layer 28, the p-side current diffusion layer 34, and the n-side current diffusion layer 36.

The protective layer 38 includes a first dielectric layer 44, a second dielectric layer 46, and a third dielectric layer 48. Each of the first dielectric layer 44, the second dielectric layer 46, and the third dielectric layer 48 is made of a

material that does not substantially absorb the deep ultra-violet light emitted by the active layer 26 and is made of a material having a transmittance equal to or more than 80% for the wavelength of the deep ultraviolet light emitted by the active layer 26. Such a material is exemplified by an oxide material such as silicon oxide (SiO_2), aluminum oxide (Al_2O_3), and hafnium oxide (HfO_2).

The first dielectric layer 44 is directly in contact with the n-type semiconductor layer 24, the active layer 26, the p-type semiconductor layer 28, the p-side current diffusion layer 34, and the n-side current diffusion layer 36. The first dielectric layer 44 is made of a first oxide material and is made of SiO_2 , Al_2O_3 , or HfO_2 . The first dielectric layer 44 is preferably made of SiO_2 . The thickness of the first dielectric layer 44 is equal to or more than 300 nm and equal to or less than 1500 nm and is, for example, about equal to more than 600 nm and equal to or less than 1000 nm.

The second dielectric layer 46 is provided on the first dielectric layer 44 and is provided to cover the entirety of the first dielectric layer 44. The second dielectric layer 46 is made of a second oxide material different from that of the first dielectric layer 44 and is made of SiO_2 , Al_2O_3 , or HfO_2 . The second dielectric layer 46 is preferably made of Al_2O_3 . By configuring the material of the second dielectric layer 46 to be different from the material of the first dielectric layer 44, pin holes that could be produced in the first dielectric layer 44 can be blocked to increase the quality of sealing. The thickness of the second dielectric layer 46 is equal to or more than 10 nm and equal to or less than 100 nm and is, for example, equal to or more than 20 nm and equal to or less than 50 nm.

The third dielectric layer 48 is provided on the second dielectric layer 46 and is provided to cover the entirety of the second dielectric layer 46. The third dielectric layer 48 is made of a third oxide material different from the second oxide material forming the second dielectric layer 46 and is preferably made of SiO_2 . By configuring the material of the third dielectric layer 48 to be different from the material of the second dielectric layer 46, pin holes that could be produced in the second dielectric layer 46 can be blocked to increase the sealing performance. The thickness of the third dielectric layer 48 is equal to or more than 10 nm and equal to or less than 100 nm and is, for example, equal to or more than 20 nm and equal to or less than 50 nm.

The p-side pad electrode 40 is provided on the p-side current diffusion layer 34 and connects to the p-side current diffusion layer 34 in the p-side pad opening 38p. The p-side pad electrode 40 is provided to block the p-side pad opening 38p and overlaps the protective layer 38 outside the p-side pad opening 38p. The p-side pad electrode 40 is electrically connected to the p-side contact electrode 30 via the p-side current diffusion layer 34.

The n-side pad electrode 42 is provided on the n-side current diffusion layer 36 and connects to the n-side current diffusion layer 36 in the n-side pad opening 38n. The n-side pad electrode 42 is provided to block the n-side pad opening 38n and overlaps the protective layer 38 outside the n-side pad opening 38n. The n-side pad electrode 42 is electrically connected to the n-side contact electrode 32 via the n-side current diffusion layer 36.

The p-side pad electrode 40 and the n-side pad electrode 42 are portions bonded when the semiconductor light-emitting element 10 is mounted on a package substrate or the like. The p-side pad electrode 40 and the n-side pad electrode 42 include, for example, a Ni/Au, Ti/Au, or Ti/Pt/Au stack structure. The thickness of each of the p-side pad electrode 40 and the n-side pad electrode 42 is equal to or

more than 100 nm and is, for example, equal to or more than 200 nm and equal to or less than 1000 nm.

FIG. 2 schematically shows a configuration of the p-side contact electrode 30 and the p-side current diffusion layer 34. The p-side contact electrode 30 is comprised of a Rh layer. The p-side current diffusion layer 34 includes a first TiN layer 50, a multilayer metal film 52, and a second TiN layer 54. The p-side current diffusion layer 34 may further include a Ti layer 56 and an Au layer 58.

The first TiN layer 50 of the p-side current diffusion layer 34 is directly in contact with the Rh layer of the p-side contact electrode 30. The multilayer metal film 52 of the p-side current diffusion layer 34 is provided on the first TiN layer 50. The second TiN layer 54 of the p-side current diffusion layer 34 is provided on the multilayer metal film 52. The first TiN layer 50 and the second TiN layer 54 are made of conductive TiN. The thickness of each of the first TiN layer 50 and the second TiN layer 54 is equal to or more than 10 nm and equal to or less than 200 nm and is, for example, equal to or more than 50 nm and equal to or less than 150 nm.

The multilayer metal film 52 of the p-side current diffusion layer 34 includes a Ti layer 52a and a Rh layer 52b. The multilayer metal film 52 may include a plurality of Ti layers 52a and a plurality of Rh layers 52b that are alternately stacked. The thickness of each of the Ti layer 52a and the Rh layer 52b is equal to or more than 10 nm and equal to or less than 200 nm and is, for example, equal to or more than 20 nm and equal to or less than 150 nm. The film density of the Rh layer 52b included in the p-side current diffusion layer 34 is equal to or more than 12.0 g/cm^3 and is, for example, equal to or more than 12.2 g/cm^3 and less than 12.5 g/cm^3 . The film density of the Rh layer included in the p-side contact electrode 30 may be larger than the film density of the Rh layer 52b. By way of one example, the Rh layer 52b included in the p-side current diffusion layer 34 has a film density of 12.37 g/cm^3 and has a reflectivity of 66.0% for ultraviolet light having a wavelength of 280 nm. The Ar concentration in the Rh layer 52b included in the p-side current diffusion layer 34 is larger than the Ar concentration in the Rh layer included in the p-side contact electrode 30. The Ar concentration in the Rh layer 52b is, for example, equal to or more than $1 \times 10^{18} / \text{cm}^3$ and equal to or less than $5 \times 10^{20} / \text{cm}^3$.

The Ti layer 56 of the p-side current diffusion layer 34 is provided on the second TiN layer 54. The thickness of the Ti layer 56 is equal to or more than 1 nm and equal to or less than 50 nm and is, for example, equal to or more than 5 nm and equal to or less than 25 nm. The Au layer 58 of the p-side current diffusion layer 34 is provided on the Ti layer 56. The thickness of the Au layer 58 is equal to or more than 100 nm and equal to or less than 500 nm and is, for example, equal to or more than 150 nm and equal to or less than 300 nm.

FIG. 3 schematically shows a configuration of the n-side contact electrode 32 and the n-side current diffusion layer 36. The n-side contact electrode 32 includes a first Ti layer 60, an Al layer 62, a second Ti layer 64, and a TiN layer 66. The n-side current diffusion layer 36 includes a first TiN layer 68, a multilayer metal film 70, and a second TiN layer 72. The n-side current diffusion layer 36 may further include a Ti layer 74 and an Au layer 76.

The first Ti layer 60 of the n-side contact electrode 32 is directly in contact with the second upper surface 24b of the n-type semiconductor layer 24. The thickness of the first Ti layer 60 is equal to or more than 1 nm and equal to or less than 10 nm and is, preferably, equal to or less than 5 nm or equal to or less than 2 nm. The Al layer 62 of the n-side

contact electrode **32** is provided on the first Ti layer **60** and is directly in contact with the first Ti layer **60**. The thickness of the Al layer **62** is equal to or more than 200 nm and is, for example, equal to or more than 300 nm and equal to or less than 1000 nm. The second Ti layer **66** of the n-side contact electrode **32** is provided on the Al layer **62** and is directly in contact with the Al layer **62**. The thickness of the second Ti layer **64** is equal to or more than 1 nm and equal to or less than 50 nm and is, for example, equal to or more than 5 nm and equal to or less than 25 nm. The TiN layer **66** of the n-side contact electrode **32** is provided on the second Ti layer **64** and is directly in contact with the second Ti layer **64**. The TiN layer **66** is made of conductive TiN. The thickness of the TiN layer **66** is equal to or more than 5 nm and equal to or less than 100 nm and is, for example, equal to or more than 10 nm and equal to or less than 50 nm.

The first TiN layer **68** of the n-side current diffusion layer **36** is directly in contact with the upper surface **32a** and the side surface **32b** of the n-side contact electrode **32**. The multilayer metal film **70** of the n-side current diffusion layer **36** is provided on the first TiN layer **68**. The second TiN layer **72** of the n-side current diffusion layer **36** is provided on the multilayer metal film **70**. The first TiN layer **68** and the second TiN layer **72** are made of conductive TiN. The thickness of each of the first TiN layer **68** and the second TiN layer **72** is equal to or more than 10 nm and equal to or less than 200 nm and is, for example, equal to or more than 50 nm and equal to or less than 150 nm.

Like the multilayer metal film **52** of the p-side current diffusion layer **34**, the multilayer metal film **70** of the n-side current diffusion layer **36** is comprised of a Ti layer **70a** and a Rh layer **70b**. The multilayer metal film **70** may include a plurality of Ti layers **70a** and a plurality of Rh layers **70b** that are alternately stacked. The thickness of each of the Ti layer **70a** and the Rh layer **70b** included in the multilayer metal film **70** is equal to or more than 10 nm and equal to or less than 200 nm and is, for example, equal to or more than 20 nm and equal to or less than 150 nm. The film density and the Ar concentration of the Rh layer **70b** included in the n-side current diffusion layer **36** are similar to the film density and the Ar concentration of the Rh layer **52b** included in the p-side current diffusion layer **34**. The film density of the Rh layer **70b** is equal to or more than 12.0 g/cm³ and is, for example, equal to or more than 12.2 g/cm³ and equal to or less than 12.5 g/cm³. The Ar concentration in the Rh layer **70b** is equal to or more than 1×10¹⁸/cm³ and less than 1×10²¹/cm³.

The Ti layer **74** of the n-side current diffusion layer **36** is provided on the second TiN layer **72**. The thickness of the Ti layer **74** is equal to or more than 1 nm and equal to or less than 50 nm and is, for example, equal to or more than 5 nm and equal to or less than 25 nm. The Au layer **76** of the n-side current diffusion layer **36** is provided on the Ti layer **74**. The thickness of the Au layer **76** is equal to or more than 100 nm and equal to or less than 500 nm and is, for example, equal to or more than 150 nm and equal to or less than 300 nm.

A description will now be given of a method of manufacturing the semiconductor light-emitting element **10**. FIGS. 4-9 schematically show steps of manufacturing the semiconductor light-emitting element **10**. First, referring to FIG. 4, the base layer **22**, the n-type semiconductor layer **24**, the active layer **26**, and the p-type semiconductor layer **28** are formed on the first principal surface **20a** of the substrate **20** sequentially.

The substrate **20** is, for example, a patterned sapphire substrate. The base layer **22** includes, for example, an HT-AlN layer and an undoped AlGaIn layer. The n-type

semiconductor layer **24**, the active layer **26**, and the p-type semiconductor layer **28** are semiconductor layers made of an AlGaIn-based semiconductor material, an AlN-based semiconductor material, or a GaN-based semiconductor material and can be formed by a well-known epitaxial growth method such as the metal organic vapor phase epitaxy (MOVPE) method and the molecular beam epitaxy (MBE) method.

Subsequently, as shown in FIG. 4, a mask **80** is formed on the upper surface **28a** of the p-type semiconductor layer **28** by using, for example, a publicly known lithographic technology. The p-type semiconductor layer **28** and the active layer **26** in a region not overlapping the mask **80** are removed by dry-etching or the like while the mask **80** is being formed, to expose the second upper surface **24b** of the n-type semiconductor layer **24**. This etching step forms the side surfaces **28b** of the p-type semiconductor layer **28**, the side surface **26b** of the active layer **26**, and the second upper surface **24b** of the n-type semiconductor layer **24**. The mask **80** is then removed.

Then, as shown in FIG. 5, the p-side contact electrode **30** is formed on the upper surface **28a** of the p-type semiconductor layer **28** by using, for example, a publicly known lithographic technology. The p-side contact electrode **30** includes a Rh layer directly in contact with the upper surface **28a** of the p-type semiconductor layer **28**. The Rh layer of the p-side contact electrode **30** is formed by deposition at a temperature equal to or less than 100° C. By forming the Rh layer by deposition, the damage to the upper surface **28a** of the p-type semiconductor layer **28** can be reduced and the contact resistance of the p-side contact electrode **30** can be improved as compared with the case of using sputtering.

After the p-side contact electrode **30** is formed, the p-side contact electrode **30** is annealed. The p-side contact electrode **30** is annealed by using, for example, the rapid thermal annealing (RTA) method at a temperature equal to or more than 500° C. and equal to or less than 650° C. The annealing process of the p-side contact electrode **30** lowers the contact resistance of the p-side contact electrode **30** and increases the film density of the Rh layer included in the p-side contact electrode **30** to be equal to or more than 12 g/cm³. The film density of the Rh layer formed by deposition at a temperature equal to or less than 100° C. is less than 12 g/cm³ and is, for example, equal to or more than 11.6 g/cm³ and equal to or less than 11.9 g/cm³. Further, the reflectivity of the Rh layer formed by deposition at a temperature equal to or less than 100° C. for the wavelength of 280 nm is less than 65% and is, for example, about 60%-61%. Meanwhile, the film density of the Rh layer of the p-side contact electrode **30** after the annealing process will be, for example, equal to or more than 12.2 g/cm³ and equal to or less than 12.5 g/cm³. The reflectivity of the Rh layer of the p-side contact electrode **30** for the wavelength of 280 nm after the annealing process is equal to or more than 65% and is, for example, 66.8%.

Then, as shown in FIG. 6, the n-side contact electrode **32** is formed on the second upper surface **24b** of the n-type semiconductor layer **24** by using, for example, a publicly known lithographic technology. The n-side contact electrode **32** is in contact with the second upper surface **24b** of the n-type semiconductor layer **24** and includes the first Ti layer **60**, the Al layer **62**, the second Ti layer **64**, and the TiN layer **66** (see FIG. 3) stacked successively. The first Ti layer **60**, the Al layer **62**, the second Ti layer **64**, and the TiN layer **66** forming the n-side contact electrode **32** are formed by sputtering.

After the n-side contact electrode **32** is formed, the n-side contact electrode **32** is annealed. The n-side contact elec-

trode 32 is annealed by using, for example, the RTA method at a temperature equal to or more than 500° C. and equal to or less than 650° C. The annealing process of the n-side contact electrode 32 lowers the contact resistance of the n-side contact electrode 32.

Subsequently, as shown in FIG. 7, the p-side current diffusion layer 34 is formed to cover the upper surface 30a and the side surface 30b of the p-side contact electrode 30, and the n-side current diffusion layer 36 is formed to cover the upper surface 32a and the side surface 32b of the n-side contact electrode 30, by using, for example, a publicly known lithographic technology. The p-side current diffusion layer 34 and the n-side current diffusion layer 36 can be formed simultaneously at a temperature equal to or less than 100° C. by sputtering that uses an Ar gas. The p-side current diffusion layer 34 and the n-side current diffusion layer 36 may be formed separately. As shown in FIGS. 2 and 3, the first TiN layers 50, 68 are formed first, and then the multilayer metal films 52, 70 including the Ti layers 52a, 70a and the Rh layers 52b, 70b are formed on the first TiN layers 50, 68, and the second TiN layers 54, 72 are formed on the multilayer metal films 52, 70. The Ti layers 56, 74 and the Au layers 58, 76 may further be formed on the second TiN layers 54, 72. The film density of the Rh layer formed by sputtering at a temperature equal to or less than 100° C. is less than 12.0 g/cm³ and is, for example, equal to or more than 11.6 g/cm³ and equal to or less than 11.9 g/cm³. The reflectivity of the Rh layer formed by sputtering at a temperature equal to or less than 100° C. for the wavelength of 280 nm is less than 65% and is, for example, about 60%-61%.

Subsequently, as shown in FIG. 8, a mask 82 is formed on the n-side semiconductor layer 24, the active layer 26, the p-type semiconductor layer 28, the p-side current diffusion layer 34, and the n-side current diffusion layer 36 by using, for example, a publicly known lithographic technology. After the mask 82 is formed, the n-type semiconductor layer 24 in a region not overlapping the mask 82 is removed by dry-etching or the like to expose the upper surface 22a of the base layer 22. This etching step forms the side surface 24c of the n-type semiconductor layer 24. The mask 82 is then removed.

Subsequently, as shown in FIG. 9, the protective layer 38 is formed to cover the entirety of the upper surface of the element structure. First, the first dielectric layer 44 made of the first oxide material is formed. The first dielectric layer 44 can be made of SiO₂ and can be formed by plasma enhanced chemical vapor deposition (PECVD). The first dielectric layer 44 is provided to cover the upper surface 22b of base layer 22, the second upper surface 24b and the side surface 24c of the n-type semiconductor layer 24, a side surface 26c of the active layer 26, the upper surface 28a and a side surface 28c of the p-type semiconductor layer 28, the p-side current diffusion layer 34, and the n-side current diffusion layer 36.

Subsequently, the second dielectric layer 46 made of the second oxide material is formed on the first dielectric layer 44. The second dielectric layer 46 is formed to cover the entirety of the upper surface of the first dielectric layer 44. The second dielectric layer 46 can be made of Al₂O₃ and can be formed by atomic layer deposition (ALD). By using the ALD method, a tight dielectric film having a high film density can be formed. The third dielectric layer 48 made of SiO₂ is then formed on the second dielectric layer 46. The third dielectric layer 48 is formed to cover the entirety of the upper surface of the second dielectric layer 46. The third dielectric layer 48 can be formed by using the ALD method.

In the step of forming the protective layer 38, the p-side current diffusion layer 34 and the n-side current diffusion layer 36 are heated at a temperature equal to or more than 200° C. and equal to or less than 400° C. The film density of the Rh layers 52b, 70b included in the p-side current diffusion layer 34 and the n-side current diffusion layer 36 is increased to be equal to or more than 12.0 g/cm³ by heating the p-side current diffusion layer 34 and the n-side current diffusion layer 36 at a temperature equal to or more than 200° C. and equal to or less than 400° C. The film density of the Rh layer formed by sputtering at a temperature equal to or less than 100° C. is less than 12.0 g/cm³ and is, for example, equal to or more than 11.6 g/cm³ and equal to or less than 11.9 g/cm³. Further, the reflectivity of the Rh layers 52b, 70b formed by sputtering at a temperature equal to or less than 100° C. for ultraviolet light having a wavelength of 280 nm is less than 65% and is, for example, about 60%-61%. Meanwhile, the Rh layers 52b, 70b heated at a temperature equal to or more than 200° C. and equal to or less than 400° C. has, for example, a film density equal to or more than 12.2 g/cm³ and less than 12.5 g/cm³ and has a reflectivity equal to or more than 65% (for example, 66%) for ultraviolet light having a wavelength of 280 nm.

When the Rh layer formed by sputtering at a temperature equal to or less than 100° C. is annealed at a temperature equal to or more than 500° C. and equal to or less than 650° C., the film density of the annealed Rh layer will be less than 12.0 g/cm³. It is therefore preferred to anneal the p-side contact electrode 30 and the n-side contact electrode 32 before the p-side current diffusion layer 34 and the n-side current diffusion layer 36 are formed.

Subsequently, as shown in FIG. 1, the protective layer 38 is removed in part by dry-etching or the like to form the p-side pad opening 38p and the n-side pad opening 38n, by using, for example, a publicly known lithographic technology. The p-side pad opening 38p and the n-side pad opening 38n are formed to extend through the first dielectric layer 44, the second dielectric layer 46, and the third dielectric layer 48. The p-side current diffusion layer 34 is exposed in the p-side pad opening 38p, and the n-side current diffusion layer 36 is exposed in the n-side pad opening 38n. Subsequently, the p-side pad electrode 40 connected to the p-side current diffusion layer 34 in the p-side pad opening 38p is formed to block the p-side pad opening 38p, and the n-side pad electrode 42 connected to the n-side current diffusion layer 36 in the n-side pad opening 38n is formed to block the n-side pad opening 38n. The p-side pad electrode 40 and the n-side pad electrode 42 may be formed simultaneously but may be formed separately.

The semiconductor light-emitting element 10 shown in FIG. 1 is completed through the steps described above.

According to the embodiment, adhesion of the protective layer 38 to the p-side contact electrode 30 can be improved by covering the upper surface 30a and the side surface 30b of the p-side contact electrode 30, which includes the Rh layer, and forming the protective layer 38 in contact with the p-side current diffusion layer 34.

According to the embodiment, the Rh layer included in the p-side current diffusion layer 34 can be used as an etching stop layer when the pad opening 38p is formed in the protective layer 38. This prevents a damage to the Rh layer included in the p-side contact electrode 30 and prevents the reflective characteristics of the p-side contact electrode 30 from being lowered.

According to the embodiment, the film density of the Rh layer included in the p-side contact electrode 30 is equal to or more than 12.0 g/cm³ so that the Rh layer included

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in the p-side contact electrode **30** can be configured to be dense to increase the reflectivity for ultraviolet light. Further, the film density of the Rh layer **52b** included in the p-side current diffusion layer **34** is also equal to or more than 12.0 g/cm^3 so that the Rh layer included in the p-side current diffusion layer **34** can be configured to be dense to increase the quality of sealing by the p-side current diffusion layer **34**.

According to the embodiment, the Rh layer included in the p-side contact electrode **30** can be configured to be dense and the reflectivity for ultraviolet light can be increased by configuring the film density of the Rh layer included in the p-side contact electrode **30** to be larger than the film density of the Rh layer **52b** included in the p-side current diffusion layer **34** and, for example, equal to or more than 12.4 g/cm^3 .

According to the embodiment, the Ar concentration in the Rh layer included in the p-side contact electrode **30** is smaller than the Ar concentration in the Rh layer **52** included in the p-side current diffusion layer **34** and is less than $1 \times 10^{18} / \text{cm}^3$ so that the film quality of the Rh layer included in the p-side contact electrode **30** can be improved. This allows the p-side contact electrode **30** to function as a highly efficient reflection electrode.

According to the embodiment, inclusion of the Rh layer **52b** in the p-side current diffusion layer **34** can increase the quality of sealing of the p-side contact electrode **30**. Similarly, inclusion of the Rh layer **70b** in the n-side current diffusion layer **36** can increase the quality of sealing of the n-side contact electrode **32**. Further, the film density of the Rh layers **52b**, **70b** included in the p-side current diffusion layer **34** and the n-side current diffusion layer **36** is equal to or more than 12.0 g/cm^3 so that the Rh layers **52b**, **70b** included in the p-side current diffusion layer **34** and the n-side current diffusion layer **36** can be configured to be dense to increase the quality of sealing even further.

According to the embodiment, formation of the Rh layer included in the p-side contact electrode **30** by deposition can suppress the damage done to the upper surface **28a** of the p-type semiconductor layer **28** when the Rh layer is formed and lower and the contact resistance of the p-side contact electrode **30** more successfully than sputtering. Meanwhile, formation of the Rh layer **52b** included in the p-side current diffusion layer **34** by sputtering can increase adhesion to the p-side contact electrode **30** and suppress exfoliation of the p-side current diffusion layer **34** more successfully than deposition.

According to the embodiment, formation of the Rh layer included in the p-side contact electrode **30** by deposition can prevent mixture of the Ar gas used in sputtering and improve the film quality of the Rh layer as compared with sputtering.

According to the embodiment, annealing the p-side contact electrode **30** at a temperature equal to or more than 500°C . and equal to or less than 650°C . can lower the contact resistance of the p-side contact electrode **30** and improve the film density of the Rh layer included in the p-side contact electrode **30** as compared to the levels before the annealing.

According to the embodiment, heating the p-side current diffusion layer **34** and the n-side current diffusion layer **36** at a temperature equal to or more than 200°C . and equal to or less than 400°C . can increase the film density of the Rh layers **52b**, **70b** included in the p-side current diffusion layer **34** and the n-side current diffusion layer **36** as compared to the level before the heating. This can increase the quality of sealing of the p-side current diffusion layer **34** and the n-side current diffusion layer **36**.

Described above is an explanation based on an exemplary embodiment. The embodiment is intended to be illustrative

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only and it will be understood by those skilled in the art that various design changes are possible and various modifications are possible and that such modifications are also within the scope of the present invention.

A description will be given below of some aspects of the present invention.

A semiconductor light-emitting element according to the first aspect includes: an n-type semiconductor layer made of an n-type AlGaIn-based semiconductor material; an active layer provided on the n-type semiconductor layer and made of an AlGaIn-based semiconductor material; a p-type semiconductor layer provided on the active layer; a p-side contact electrode that includes a Rh layer in contact with an upper surface of the p-type semiconductor layer; and a p-side current diffusion layer that is in contact with an upper surface and a side surface of the p-side contact electrode and includes a TiN layer, a Ti layer, a Rh layer, and a TiN layer stacked successively, wherein an Ar concentration in the Rh layer included in the p-side contact electrode is smaller than an Ar concentration in the Rh layer included in the p-side current diffusion layer. According to the first aspect, the quality of sealing of the p-side contact electrode can be increased by covering the upper surface and the side surface of the p-side contact electrode, which includes the Rh layer, by the p-side current diffusion layer, which includes the Rh layer. Further, the film quality of the p-side contact electrode can be improved, and the p-side contact electrode can be allowed to function as a highly efficient reflection electrode by configuring the Ar concentration to be small.

The second aspect of the present invention relates to the semiconductor light-emitting element according to the first aspect, wherein the Ar concentration in the Rh layer included in the p-side contact electrode is less than $1 \times 10^{18} / \text{cm}^3$, and the Ar concentration in the Rh layer included in the p-side current diffusion layer is equal to or more than $1 \times 10^{18} / \text{cm}^3$. According to the second aspect, the film quality of the p-side contact electrode can be improved, and the p-side contact electrode can be allowed to function as a highly efficient reflection electrode.

What is claimed is:

1. A semiconductor light-emitting element comprising:
 - an n-type semiconductor layer made of an n-type AlGaIn-based semiconductor material;
 - an active layer provided on the n-type semiconductor layer and made of an AlGaIn-based semiconductor material;
 - a p-type semiconductor layer provided on the active layer;
 - a p-side contact electrode that includes a Rh layer in contact with an upper surface of the p-type semiconductor layer; and
 - a p-side current diffusion layer that is in contact with an upper surface and a side surface of the p-side contact electrode and includes a TiN layer, a Ti layer, a Rh layer, and a TiN layer stacked successively, wherein
 - an Ar concentration in the Rh layer included in the p-side contact electrode is smaller than an Ar concentration in the Rh layer included in the p-side current diffusion layer.
2. The semiconductor light-emitting element according to claim 1, wherein
 - the Ar concentration in the Rh layer included in the p-side contact electrode is less than $1 \times 10^{18} / \text{cm}^3$, and
 - the Ar concentration in the Rh layer included in the p-side current diffusion layer is equal to or more than $1 \times 10^{18} / \text{cm}^3$.

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